

Skills Summary

Composing Piecewise and Nonlinear Functions

Use this reference while completing your worksheet or when studying.

Definition. $(f \circ g)(x) = f(g(x))$ — the **inner** function g acts first, and its output becomes the input of f .

Two rules that cause most errors.

- Order matters: $f \circ g$ and $g \circ f$ are different functions.
- Domain: the input must be legal for g , and $g(x)$ must be legal for f . Simplifying can hide a restriction, so read the domain off the composition before simplifying.

Examples on this card use $a(x) = 3x - 1$, $b(x) = x^2 + 2$, and $w(x) = \begin{cases} -x, & x < 0 \\ x + 6, & x \geq 0 \end{cases}$

1. Evaluating a composition at a number

Work **inside out**. Compute the inner value, write it down, then substitute that single number into the outer function. Never substitute into both at once.

Example: Find $(a \circ b)(2)$.

Step 1. The composition names b as the inner function, so the only legal first move is to evaluate b at 2 and carry that output outward.

Step 2. $b(2) = 2^2 + 2 = 6$, and then $a(6) = 3(6) - 1$.

$$(a \circ b)(2) = 17$$

Try it: Find $(b \circ a)(2)$ — the same two functions, other order.

check: 27

2. Simplifying a composition into one rule

Substitute the whole inner *rule* into the outer function's variable slot, keeping it in brackets, then expand: $a(b(x)) = 3(b(x)) - 1$. Expand squares as $(mx + c)^2 = m^2x^2 + 2mcx + c^2$ — never as $m^2x^2 + c^2$.

Example: Write $(a \circ b)(x)$ as one simplified rule.

Step 1. A rule, not a number, is wanted, so the inner rule goes into the slot where a 's variable stands and nothing is evaluated yet.

Step 2. $a(b(x)) = 3(x^2 + 2) - 1 = 3x^2 + 6 - 1$.

$$3x^2 + 5 \quad (\text{check at } x = 2 : 12 + 5 = 17)$$

Try it: Write $(b \circ a)(x)$ as one simplified rule.

check: $x^3 + 9x - 6$

3. Piecewise outer function: pick the piece

A piecewise function tests **its own input**. In $w(g(x))$ that input is $g(x)$, *not* x — so compute the inner value first, compare *that* with the break point, and only then choose a rule. For a piecewise rule in x , translate the break: w breaks at 0, so $w(a(x))$ breaks where $a(x) = 0$.

Example: Find $w(a(0))$.

Step 1. The piece is chosen by what reaches w , so evaluate the inner function before looking at either condition.

Step 2. $a(0) = 3(0) - 1 = -1$. Since $-1 < 0$, w uses the $-x$ piece, giving $-(-1)$.

$$w(a(0)) = 1$$

Try it: Find $w(a(2))$. (Check the sign of $a(2)$ before choosing.)

check: 11

4. Solving a composition equation

Simplify the composition to one rule, set it equal to the target, and solve. Quadratic compositions usually have **two** solutions — take both roots.

With a piecewise outer function, solve *each* piece and then test every candidate against the condition that selects that piece; a candidate outside its own region is rejected.

Example: Solve $(a \circ b)(x) = 32$.

Step 1. One rule for the composition already exists from section 2, so the equation can be solved directly instead of unwinding two functions.

Step 2. $3x^2 + 5 = 32$ gives $3x^2 = 27$, so $x^2 = 9$, which has two real roots.

$$x = -3 \text{ or } x = 3$$

Try it: Solve $(b \circ a)(x) = 3$, using the rule from section 2.

check: $\frac{x}{2} = x$ to $0 = x$